

Data Quality Audit Report

Superstore Dataset | Task 24 (Level 2, Day 24)

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Track	Data Analytics
Tools used	Python, Pandas (Google Colab)
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Dataset	superstore.csv (10,800 rows x 21 columns, as received)

Executive Summary

The Superstore dataset was audited for four categories of quality issues: **missing values, duplicates, range violations and consistency problems**. The file as received contained **10,800 rows**, of which **806 (7.46%)** were completely blank apart from their IDs, leaving **9,994 genuine records**. Beyond these, only 11 missing Postal Codes and 1 duplicate record were found. All range and consistency checks passed. After cleaning, the final dataset contains **9,993 rows** and no missing values.

Rows received	Blank rows removed	Duplicates removed	Final clean rows
10,800	806	1	9,993

1. Objective and Scope

The objective of this task is to audit a dataset for missing, duplicate, range and consistency issues, and to produce a **repeatable quality checklist** that can be applied to any tabular dataset. The audit covered all 21 columns of the Superstore dataset: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region, Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount and Profit.

2. Methodology

- Loaded the CSV into a Pandas DataFrame and reviewed structure with `info()` and `describe()`.
- Calculated missing counts and percentages per column.
- Checked duplicates on full rows and again ignoring Row ID (to catch repeated records with different IDs).
- Applied range rules to Sales, Quantity, Discount and the two date columns.
- Checked consistency of categorical values, whitespace, Customer ID to Name mapping, and Sub-Category to Category mapping.
- Quantified every issue (count and percentage), logged it row by row, then cleaned the data.

3. Validation Rules Checklist (Repeatable)

The rules below form the reusable checklist. Each rule is a yes/no test that can be run on any similar dataset.

#	Dimension	Rule	Result
1	Missing	No row should have all data columns empty	FAIL (806 rows)
2	Missing	Critical fields (IDs, dates, Sales, Quantity) must not be null	PASS after blank-row removal
3	Missing	Postal Code should be present for every record	FAIL (11 rows)
4	Duplicate	No fully identical rows	PASS (0)
5	Duplicate	No identical records differing only by Row ID	FAIL (1 row)
6	Range	Sales > 0	PASS (min 0.444)
7	Range	Quantity > 0	PASS (min 1)
8	Range	Discount between 0 and 1	PASS (0.00 to 0.80)
9	Range	Ship Date >= Order Date	PASS (0 violations)
10	Range	Dates parse to valid dates	PASS (0 invalid)
11	Consistency	Categorical values have no spelling or case variants	PASS
12	Consistency	No leading or trailing whitespace in text fields	PASS
13	Consistency	One Customer ID maps to one Customer Name	PASS (0 conflicts)
14	Consistency	Each Sub-Category belongs to one Category	PASS (0 conflicts)

4. Findings

Percentages are calculated on the original 10,800 rows unless stated otherwise.

Issue	Column(s)	Count	% of rows	Severity	Action taken
Fully blank rows	All except Row ID and Order ID	806	7.46%	High	Removed
Missing Postal Code	Postal Code	11	0.10%	Low	Filled with 05401
Duplicate record (ignoring Row ID)	All except Row ID	1	0.01%	Medium	Removed (first kept)
Range violations	Sales, Quantity, Discount, Dates	0	0.00%	Passed	None
Consistency issues	Categorical, IDs	0	0.00%	Passed	None
Negative profit orders	Profit	1,870	17.31%*	Info	Retained

*On the final 9,993 clean rows, negative profit orders represent 18.71%.

4.1 Fully blank rows (High)

806 rows had a Row ID and Order ID but every other column was empty. Every column showed exactly 806 missing values, which indicated whole blank rows rather than scattered gaps. These rows sit after the 9,994th record. Imputing them would have created fake records, so they were removed.

4.2 Missing Postal Code (Low)

After removing blank rows, 11 records still had no Postal Code. All 11 belonged to Burlington, Vermont, so they were filled with 05401, the postal code for that city.

4.3 Duplicate record (Medium)

No fully identical rows existed, but one pair of records was identical in every column except Row ID. It was treated as a probable duplicate entry and the first occurrence was kept.

4.4 Observations (not errors)

- **Negative profit:** 1,870 orders are loss-making (minimum profit -6,599.98). These are valid business outcomes, so they were retained and flagged for analysis.
- **Large values:** Sales has a median of 54.49 but a maximum of 22,638.48. These look like genuine bulk orders and were not removed.
- **Date coverage:** Orders span 2015-01-03 to 2018-12-30 with no invalid or out-of-sequence dates.

5. Summary Statistics

Column	Min	Median	Mean	Max
Sales	0.44	54.49	229.86	22,638.48
Quantity	1	3	3.79	14
Discount	0.00	0.20	0.16	0.80
Profit	-6,599.98	8.67	28.66	8,399.98

Statistics shown are from the 9,994-row check before the single duplicate was removed; the difference is negligible.

6. Cleaning Steps Applied

1. Identified rows where all data columns were null (Row ID and Order ID excluded) and removed them: 10,800 to 9,994 rows.
2. Filled 11 missing Postal Codes for Burlington, Vermont with 05401.
3. Removed 1 duplicate record (ignoring Row ID), keeping the first occurrence: 9,994 to 9,993 rows.
4. Parsed Order Date and Ship Date using the month/day/year format and confirmed validity.
5. Saved the outputs: cleaned dataset, audit summary and row-level issue log.

7. Recommendations

- **Fix at source:** investigate why 806 blank rows were appended to the export and prevent them in the data pipeline.
- **Make Postal Code mandatory** at data entry, and store it as text to preserve leading zeros (for example 05401).
- **Add a uniqueness check** on business keys (Order ID + Product ID + Customer ID) rather than relying on Row ID alone.
- **Automate the checklist:** run the 14 rules above on every new data load and review the summary before analysis.
- **Review loss-making orders** with high discounts, since they account for roughly 18.7% of records.

8. Deliverables

File	Description
Task24_Data_Quality_Audit_Report.pdf	This audit report
issue_log.csv	Row-level log of every issue with column, severity and action
audit_summary.csv	Issue counts, percentages, severity and action
cleaned_sample.csv	Cleaned dataset (9,993 rows x 21 columns)
Notebook (.ipynb)	Google Colab notebook with the full audit code

9. Conclusion

The Superstore dataset is structurally sound once the 806 blank rows are removed. Range and consistency quality is high, with no violations found. The only data repairs needed were 11 Postal Codes and 1 duplicate. The checklist in Section 3 can now be reused on other datasets to make future audits quick and consistent.